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Incorporating pragmatism in a behaviour change-led climate adaptation project: a collaborative reflection

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Abstract

Purpose – This article argues the value of integrating pragmatism in applying behavioural science to complex challenges. We describe a behaviour change-led knowledge co-production process in the specific context of climate change in Australia. This process was led by an interdisciplinary research team who struggled with the limitations of the prevailing deterministic behaviour change paradigms, such as the "test, learn, adapt" model, which often focuses narrowly on individual behaviours and fails to integrate multiple interpretations from diverse stakeholders into their knowledge co-production process.

Design/methodology/approach – This article uses collaborative reflection as a method of inquiry. We document the team's experience of a recent challenge-led, programatic research initiative that applied behaviour change strategies to reduce climate vulnerabilities. We demonstrate the necessity of critical reflection and abductive reasoning in the face of the complexities inherent in knowledge co-production addressing complex problems. It underscores the importance of accommodating diverse perspectives and contextual nuances over a one-size-fits-all method.

Findings – The article shares lessons learnt about integrating collaborative and critical reflection throughout a project cycle and demonstrates the capacity of abductive reasoning to ease the challenges arising from the tension between behaviour change paradigms and knowledge co-production principles. This approach allows for a more adaptable and context-sensitive application, acknowledging the multiplicity of understandings and the dynamic nature of behavioural change in relation to climate adaptation.

Originality/value – This reflection contributes original insights into the fusion of pragmatism with behaviour change strategies, proposing a novel framework that prioritises flexibility, context-specificity and the recognition of various stakeholder perspectives in the co-production of knowledge.

Keywords Abductive reasoning, Behaviour change, Climate adaptation, Critical reflection, Knowledge co-production, Pragmatism, Qualitative research

Paper type Case study

Introduction

Behaviour change approaches have been applied to address many complex problems, including sustainable transitions, environmental degradation, and climate change (Kaufman *et al.*, 2021). While these approaches are helpful, their foundations – traditional behavioural science – have been sometimes criticised for their narrow focus on the individual and behaviour. For sociologists, social structures play an important role in shaping agency and actors' behaviours (see, for example, Bourdieu, 1977, 1990; Giddens, 1990). Utilising social

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practice theories, sociologists like Shove (2010) have criticised the limitations of the individualistic approach in solving large-scale problems such as climate change. In her words: "framing the problem of climate change as a problem of human behaviour marginalises and in many ways excludes serious engagement with other possible analyses, including those grounded in social theories of practice and transition" (Shove, 2010, p. 1274).

Recent critiques have also highlighed the behaviour change approach's limitations, including its reliance on the traditional behavioural science's "test, learn, adapt" method (Hansen, 2018), overemphasis on randomised control trials (RCTs), challenges in replicating findings (Bryan *et al.*, 2021), and individualistic focus (Chater and Loewenstein, 2022; Ewert, 2020). This approach focuses too narrowly on limited impact, fails to scale up effectively, relies on flawed evidence, lacks precision, displays overconfidence, and promotes a control paradigm that disregards participants' own objectives and plans (Hallsworth, 2023). One example of this approach is assuming a linear, step-by-step process that ignores indirect impacts and unintended consequences, which can only be captured by in-depth information gathered from diverse participants and stakeholders. This is potentially problematic in addressing even relatively stable, well understood problems in public policy and administration; and raises legitimate questions about behavioural sciences applicability in addressing urgent systemic transitions that involve multiple actors, multiple levels, and multiple facets (Kaufman *et al.*, 2021; Boulet *et al.*, 2021, 2022). From a sociological perspective, individuals often rely on their socio-cultural traditions and norms to interpret the challenges they face and subsequently inform their decision-making and action (Lash, 1994; Nguyen-Trung, 2023). As such, it is of importance to uncover and incorporate these multiple interpretations in understanding a project's context and problems before designing behaviour change interventions. Additionally, individualistic and/or deterministic behaviour science paradigms are in conflict with knowledge co-production and participatory approaches. While the behaviour change lens adopts a more individualistic approach, emphasising deterministic and theory-driven orientations, the practice of knowledge co-production tends to be more open-ended and data-driven. Norström *et al.* (2020) identified four principles for driving high-quality knowledge co-production processes in sustainability: (1) context-based; (2) pluralistic; (3) goal-oriented; and (4) interactive. These principles emphasise the need for co-production processes to be situated within specific social, economic, and ecological contexts, to involve multiple perspectives and ways of knowing, to have well-defined goals that are shared amongst participants, and to encourage continual reflection, learning, and adaptive management. Bryan *et al.* (2021) suggest that a "heterogeneity revolution" - recognising the diverse nature of treatment effects and the expected variation in effect estimates across studies - as a solution to the crisis of confidence in the replicability of traditional behavioural science findings in policymaking. However, their singular focus on causal mechanisms and generalisation does not align with the need for co-design collaboration, which requires multiple explanations for the challenges at hand.

This article presents a case study of the Climate Adaptation Mission in Australia, a behavioural science-led project conducted from 2021 to 2023 that substantially involved knowledge co-production in the realm of climate change adaptation. Our research team was interdisciplinary, consisting of researchers from sociology, behavioural science, human ecology, and social psychology. Guided by a behaviour change approach, the project faced challenges concerning the integration of behavioural science in the context of co-producing knowledge, particularly with the participation of multiple stakeholders, including those from Traditional Owners (namely, indigenous communities), land managers, and many layers of governance in Australia. As pioneers in applying behavioural science for positive social and environmental change since 2011, we at BehaviourWorks Australia believe it is our responsibility to challenge and innovate our traditional behaviour change approaches, making them more relevant in the context of escalating climate change and disaster risks. In

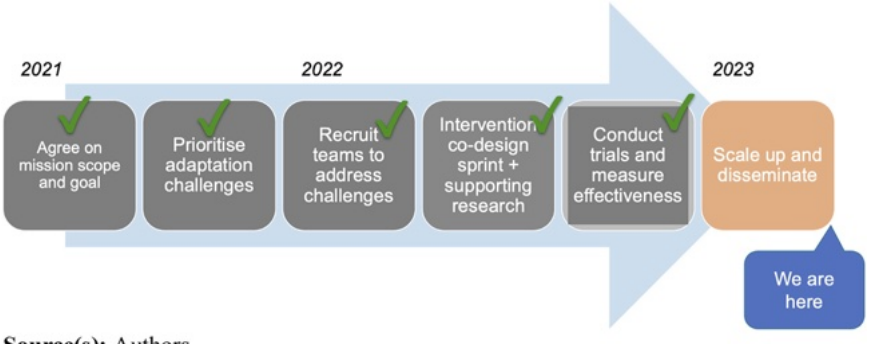
this attempt, we see that collaborative and critical reflection is essential for uncovering our preconceived notions and biases, stimulating our critical thinking, and transforming our routines.

In this article, consonant with some previous sociological approaches (Shove, 2010) and recent proposals for advancing behavioural science in public policy (Ewert, 2020), we advocate for the incorporation of pragmatism in making behaviour change approaches relevant to the knowledge co-production process. Rooted in the pioneering works of pragmatists such as Charles Sanders Peirce, William James, and John Dewey, pragmatism is considered an alternative to post-positivism/idealism and social constructivism/realism, with a focus on problem-solving (Morgan, 2014). Pragmatism hinges on the production of actionable knowledge, recognises the interplay between experience, knowing, and acting, and views inquiry as an experiential process (Kelly and Cordeiro, 2020). We are not the first to advocate for the application of pragmatism to behavioural science; Cash *et al.* (2017) proposed using a pragmatic approach to enhance creativity in behavioural design, while Brierley (2017) argues that the use of pragmatism helps overcome limitations of ontological and epistemological issues. In this article, we used a collaborative and critical reflection approach to document our application of pragmatism in facilitating behavioural science-led knowledge co-production process within the Climate Adaptation Mission from 2021 to 2023.

The case of climate adaptation mission in Australia

The Climate Adaptation Mission is a program designed to implement systemic behavioural science interventions within a mission-led research and innovation framework (Mazzucato, 2018). It is an initiative of the BehaviourWorks Australia (BWA) consortium, a collaboration of a behavioural science unit under Monash Sustainable Development Institute, Monash University, Australia; and its partners, including representatives from the Australian government, the Victorian government, private companies, and other relevant agencies. The Mission aims to enhance the adaptive capacity of communities most vulnerable to climate change impacts by 2030, aligned with the UN Sustainable Development Goals. Adaptive capacity is conceptualised as the behavioural, technological, and resource adjustments necessary for thriving amidst climate change and variability.

The Mission unfolded over six stages from June 2021 to July 2023 (see Figure 1). Stage 1 (2021) involved dialogues with partners informed by a literature and practice review to finalise the Mission scope and objectives (Kaufman *et al.*, 2021). Stage 2 (late 2021) scoped the adaptation challenges facing Australia and identified a long list of challenges, which were



Source(s): Authors

Figure 1.
The climate adaptation mission process

subsequently prioritised to select 4–6 challenges for the behaviour change intervention co-design phase (Grundy *et al.*, 2021). Stage 3 (January to April 2022) assembled teams of stakeholders pertinent to the identified challenges for the co-design phase. In Stage 4 (April to August 2022), two climate adaptation interventions were co-designed via five main workshops and eight smaller team sessions. Two final challenges and two target groups were selected, whose behaviours needed to change to address the chosen challenges. Group 1 evaluated a local initiative aiming to boost farmers' engagement in natural capital market transactions in South Queensland and New South Wales. Group 2 assisted DJAARA (Dja Dja Wurrung Clans Aboriginal Corporation) in organising an adaptation forum in Central Victoria with local government climate adaptation planners, aiming to foster partnerships for climate action plans. Stage 5 (September 2022 to April 2023) centred on executing and evaluating the two climate adaptation interventions. Stage 6 (April to July 2023) seeks to scale up and disseminate the interventions to diverse settings and knowledge users.

Reflection, pragmatism and knowledge co-production

This section aims to argue for the integration of critical reflection and abductive reasoning in a behaviour change project by stating our position regarding reflection and pragmatism. Reflection is often considered part of the research process, as it enables researchers to delve deeper into data and interpretation while avoiding biases (Braun and Clarke, 2019a, b). Such reflexivity is essential for awakening our awareness (Bourdieu, 1977) and questioning unexamined assumptions (Adkins, 2003). More importantly, reflection is a "key element of problem-solving" and "the core of the learning process" because in its "critical" form, reflection empowers researchers to pose inquiries regarding their purposes, challenge their taken-for-granted assumptions, uncover hidden agendas and beliefs, and drive changes (Reynolds, 1998, pp.183-184). In this sense, reflection can also be considered a research method in its own right (Fook and Gardner, 2007, 2012). In our case, critical reflection served as a method of inquiry to enhance and transform our professional practice when conflicts arose between the requirements of knowledge co-production (i.e. pluralistic and context focused) and the limitations of the behaviour change approach (i.e. individual focused). According to Fook and Gardner (2007, p. 28), "to be reflexive, we need to be aware of the many and varied ways in which we might create, or at least influence, the type of knowledge we use." Moreover, critical reflection should function to improve the way we live and relate as human beings and, in doing so, should also refine our professional practice (Fook and Gardner, 2012). In other words, critical reflection should focus on two goals: (1) unearthing our implicit assumptions in our daily professional practice and (2) transforming our professional practice.

This critical reflection approach is harmonious with the principles of pragmatism, which champions intersubjectivity and practical knowledge. Intersubjectivity refers to the approach that flexibly incorporates both objective knowledge (i.e. there is a single "real world" out there) and subjective perspectives (i.e. every single participant has their own interpretation of reality based on their subculture, experience and personal values) in the research process (Morgan, 2007). Extreme positivist and constructivist approaches, on the other hand, have limitations in that they prioritise either absolute objectivity or absolute subjectivity, respectively. This "either-or" approach has the potential to exclude scientific understandings based on evolving research literature or participants' everyday knowledge from the decision-making processes of action research. In reality, intersubjectivity allows researchers "to work back and forth between various frames of reference" (Morgan, 2007, p. 71) to adapt to the practical contexts and stakeholders of the problem under study. This approach allows practical researchers, whose aim is to drive social and behaviour changes, to involve stakeholders and research participants in achieving a mutual understanding of the problem, creating a shared plan for action, and achieving common goals (Kelly and Cordeiro, 2020).

This pragmatic approach is relevant to the knowledge co-production process, which underscores the necessity of pluralistic thinking, which involves acknowledging multiple interpretations of reality (Norström *et al.*, 2020). In our approach, we argue that the incorporation of critical reflection into a behaviour change approach can help facilitate intersubjectivity, therefore, enhance the effectiveness of co-producing knowledge amongst stakeholders. This form of critical reflection could be conceptualised as "Nudge Plus", a term introduced by John and Stoker (2019) to extend the "reflective capacities" often overlooked in traditional nudge approaches (Thaler and Sunstein, 2008). Instead of solely motivating the public—the typical target of behavioural interventions—our critical reflection framework posits that behaviour change researchers also need space for introspection. This space enables scientists to develop intersubjectivity with co-producers, encompassing both their own reflections and those of their research participants. This balanced approach moves away from the tyranny of experts' singular truths, encouraging the acceptance of multiple epistemologies and interpretations throughout the co-production process. It fosters a respectful environment where scientists and co-producers can challenge their assumptions, resolve conflicts, and ultimately reach a mutual understanding of the problem at hand. The next section details the incorporation of reflection in our behaviour change project to facilitate intersubjectivity or shared understanding.

The authors of this article are three of the six researchers deeply involved in the Mission. The second author (behavioural scientist) and the third author (human ecologist) have co-led the Mission since its inception in June 2021. The first author (sociologist) joined the Mission in March 2022 and has significantly contributed to the co-design workshops and the Mission's interventions. As an interdisciplinary team, we endeavoured to incorporate pragmatism into our behaviour change approach, aiming to enhance its suitability and effectiveness in driving the knowledge co-production process. This involved a broad array of stakeholders, ranging from farmers, Traditional Owner communities, Aboriginal clan corporations, water authorities, a farmer cooperative, community based natural resource management groups, research institutions, and policymakers and administrators from local, state and federal government, and natural resource management regional bodies.

Ethical clearance was sought and granted prior to any data collection involving human subjects. As this article solely analyses our own reflections and project documents, ethical clearance was deemed inapplicable. As described below, reflection was considered as a method of inquiry which helped us reflect on and critique our assumptions in applying behaviour change principles and techniques in our project. All the team sessions were video recorded via Zoom (Zoom Video, 2024) and team reflexive notes were written, stored, developed, refined, and virtually analysed using Miro Board, an online collaborative platform (Miro, 2024). Miro Board is an alternative option to in-person workshops that use creative writing activities including the use of images, artefacts to elicit co-researchers' creativity (Lupton and Watson, 2021). In Miro Board, co-researchers can create different templates using graphical tools such as mapping, diagraming, or image elicitation, to encourage the interaction and creativity. We incorporated Morgan's pragmatic approach (2014, 2020) and Fook and Gardner's (2007) reflection approach in analysing our own reflection. Our analysis generally adhered to Braun and Clarke's (2006, 2022) thematic analysis approach, with a particular emphasis on attaining a mutual understanding of how to improve behaviour change interventions. According to Morgan (2007), in pragmatism, finding shared understanding or intersubjectivity is crucial in identifying and revising our beliefs through action in order to improve the consequences of our project design/interventions.

Incorporating critical reflection into a behavioural change approach to facilitate intersubjectivity

In accordance with intersubjectivity (Morgan, 2007) and the method of critical reflection proposed by Fook and Gardner (2007), we developed a process of critical reflection that serves as a transforming process for our behaviour change approach. This process consists of three stages (Figure 2).

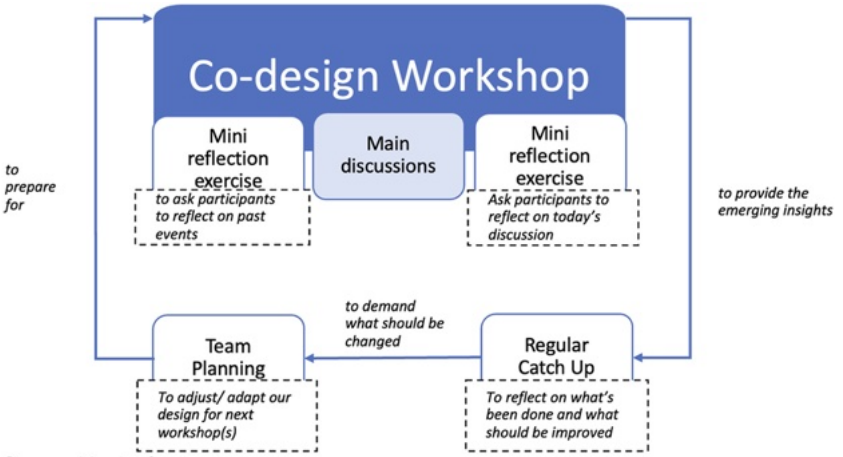
Stage 1: reflection in action

This stage involved regular meetings held in conjunction with the mission's co-design workshops, intervention implementation, and monitoring activities. The objective of this stage was to monitor progress, identify any emerging issues, and troubleshoot them in a timely manner. To achieve this, we organised small, facilitated group meetings, which involved regular catch-up sessions and team planning amongst our research team, with one member designated as a facilitator and another as a note-taker. Due to the COVID-19 pandemic, we used a hybrid mode of interaction, including face-to-face discussions whenever possible and virtual meetings via Zoom, an online video calling and meeting platform. We utilised Miro Board, an online collaborative platform, which facilitated visualising plans, taking notes, and exchanging ideas during our discussions. In each team session, we identified a specific set of questions or designed graphic templates to elicit participants' reaction and taking notes in response to these questions. The outputs from this stage included regular catch-up meeting minutes that contained descriptive reflections on progress and diagnostic reflections on emerging issues, as well as visualised models and frames developed using the Miro Board. The reflection in action stage was particularly designed to support our co-designing practice (see Figure 3), which was crucial to our knowledge co-production. The co-design process included five 3.5-h main workshops and eight 1-h team sessions in between the main workshops. Each of the main workshops involved the whole team of approximately 20 people, while each team session involved about 10 people each, working on two particular challenges, namely, farmers' engagement in natural capital market transactions and incorporation of Traditional Owners' knowledge into local councils' climate adaptation planning. We maintained a reflection process that helped set up the learning loop, as seen in Figure 3. This learning loop comprised three parts: the co-design workshops, where participants co-designed the behavioural interventions; our weekly catch-up sessions (1 h); and a visualisation planning session (1–2 h).



Figure 2.
Stages of critical reflective practice in project development

Source(s): Authors



Source(s): Authors

Figure 3.
Integrating reflection in co-design workshop stages

During each workshop, we had two reflection exercises, the first at the beginning and the second at the end. The first exercise asked participants, including our research team members, to reflect on the following questions: (1) What had been done in past events? (2) What is coming up (i.e. the concerns they had); and (3) What is needed to be changed to make our direction more appropriate (i.e. next actions and troubleshooting methods). The second exercise asked participants to reflect on what had been done during the current workshop and what key takeaways they had. These two mini-reflection exercises helped close the workshop's reflective loop and were facilitated in a focus group discussion style with the support of Miro Board (visually) and Zoom (verbally).

According Morgan (2020), the process of pragmatic inquiry involves a cycle that connects actions with beliefs and consequences. In our reflection-in-action process, we repeated this cycle by "exploring" our assumptions and beliefs through workshops, regular catch-up meetings and team planning sessions, and then "revising" these beliefs by applying them into next workshops (new action). For instance, we understood that the discussion outputs (including opinion, experiences, stories, instances, etc.) during the workshops were raw and needed to be further processed for better use. To utilise the outputs, we had two internal sessions: regular catch-ups and team planning sessions. A regular catch-up was organised every week for 1 h amongst our research team. In these meetings, each team member had a chance to reflect on what had been done, what assumptions had emerged, what was coming up, what issues needed attention, timeline checks, and next actions. Next, team planning sessions were organised before each workshop, in which we researchers spent time turning those revised beliefs into specific questions, templates, and activities (i.e. open-ended focus group discussion exercises) in Miro Board for the next workshops. These sessions required our team members to reflect on the lessons learnt from the previous workshops and, more importantly, what we should change to make the next workshop better address present issues and incorporate participants' concerns. Then every new workshop was a space for reflecting on identified or revised beliefs and assumptions. Discussions based on this new understanding would inform us on how to better identify the influences of climate challenges in Australia, structural and individual drivers of behaviour changes, or better involve relevant stakeholders in co-designing behaviour change interventions.

Stage 2: reflection on knowledge co-production

This stage took place from early December 2022 to March 2023, following the completion of our two behaviour change interventions. The objective of this stage was to reflect on the co-produced knowledge to date (from before to after the implementation of the two interventions) and identify the primary, next, and wider users of this knowledge within the mission. We conducted two activities for this stage: (1) a small facilitated group within the research team to brainstorm the knowledge co-production process and propose questions and activities for broader engagement with our partners; and (2) an expanded facilitated group where we invited partners from the Mission's steering committee and potential knowledge users to reflect on the knowledge co-production process and discuss who should be the primary, next, and wider users of the co-produced knowledge and how we should engage or communicate with them. Both group discussions were conducted via Zoom, with the use of a Miro Board as a creative space for discussion and reflection. The outputs from this stage are a collection of researchers "and partners" reflections on the knowledge co-production process and the knowledge users.

Stage 3: critical reflection on the role of behavioural change in knowledge co-production

This stage took place in February and March 2023, right before our scaling up workshops which aimed to share lessons learnt with different knowledge users of the project. Our objectives were to challenge our assumptions about the use of behaviour change approaches in the knowledge co-production process and transform our practice in the Mission's next stage. The activities for this stage included: (1) two 2-h online group discussions and two 1-h team in-person discussions where the team discussed with each other with a facilitator. We used Zoom (for online sessions) and mobile phone (for in-person sessions) to audio/video record our reflection and Miro Board to store our collaborative and personal notes and graphic models; (2) self-reflection where we asked each member to work on Miro Board to reflect on the usefulness of using a pragmatic approach in co-producing knowledge for the Mission (see Box 1); and (3) one collaborative reflection session with three researchers reflecting on how to meaningfully integrate all lessons learnt for the use of the Mission and what changes should be made in our approach. In addition to the use of Miro Board as a reflexive journal (Braun and Clarke, 2021), we also used this platform to design creative graphics, system mapping, matrices, tables, or diagrams to provoke members' reflection (Lupton and Watson, 2021). These qualitative data were used to achieve our intersubjectivity or shared understanding of the challenges our behaviour change interventions were facing and how to transform our approaches.

In sum, intersubjectivity was fostered through reflective activities such as group discussions, brainstorming sessions, and reflective writing. By integrating collaborative reflection into our co-design process, we deepened our understanding of the complex issues we faced, thereby co-creating more responsive interventions. Throughout the five co-design

Box 1. Three reflective questions

(1) What is helpful or not helpful about our behaviour change approach to facilitating the Mission's knowledge co-production process?
(2) How can we integrate the lessons learnt from that practise in a meaningful way for the sake of the Climate Mission's next stage, "Scale-up and Dissemination"?
(3) How can we improve our behaviour change approach to better address climate adaptation challenges in the future?

workshops, eight team sessions, and subsequent intervention implementation and monitoring, our collaborative stakeholder sessions were structured to be both reflective and interactive. We advocate the importance of pausing, stepping back, and engaging in reflection as a mechanism to achieve balance. This need for critical reflection stems from the risk of being unwittingly guided by our own entrenched habits or subconscious biases, particularly when applying behavioural science to the co-design and execution of interventions. Failing to reflect on our experiences could hinder our ability to learn from them and refine our methodologies for future initiatives.

Using abductive reasoning in behavioural science-led knowledge co-production

There is often a sharp distinction between post-positivism, which is often the philosophical foundation for quantitative research, and constructivism for qualitative research. Each worldview relies on different epistemological assumptions (i.e. how we know what we know) that use different types of reasoning about the connection between theory and data. While the qualitative approach often relies on induction (data-driven), the quantitative approach and behaviour change approach often rely on deduction (theory-driven). However, in the case of "wicked problems" that involve multiple structural and multifaceted forces and unknown cause-and-effect relationships (Rittel and Webber, 1973) like climate change, theory-driven approach with pre-determined frameworks may not always be effective in understanding the challenge at hand. Similar to the call for adopting "political epistemologies" that acknowledge the multiple and potentially contested ways in which science and politics interact and the various definitions of rationality (Straβheim, 2020) or the call for adopting a cultural evolutionary perspective to overcome the Western, Educated, Industrialised, Rich and Democratic societies (WEIRD) people (Schimmelpfennig and Muthukrishna, 2023), we argue that behaviour change approach needs to incorporate the principles of participatory methodologies (Norström *et al.*, 2020). These methodologies recognise the co-existence of multiple realities and different types of knowledge that affect decision- and policy-making processes. For instance, when working with a particular group, it is necessary to account for how they often rely on their unique shared epistemology (i.e. ways of knowing) rooted in their culture (Lash, 1994, 2003) and their "sense of the game" or situated reality to make practical decisions without using as much consciousness or rationality as us scientists often do (Lamaison and Bourdieu, 1986; Nguyen-Trung, 2022).

A pragmatic approach can help address these challenges. The key to pragmatism is its abductive reasoning, which allows us to move beyond the limitations of inductive and deductive extremes and focus on solving problems as they arise. This is crucial since the process of applying behaviour change on a large scale often involves many groups of stakeholders, who will co-produce different types of knowledge from different angles and methodological backgrounds. Abduction offers problem-oriented flexibility and the capability to use both theory-driven and data-driven approaches, both quantitative and qualitative methods, to suit the needs of achieving shared understanding amongst stakeholders (see Ewert, 2020). With a pragmatic approach, behaviour change researchers can still use their pre-determined theories or frameworks to support the early setup of the co-design process, but they should then be relatively open to insights from participants. Participants involved in the co-design process contribute their insights based on different cultures, knowledge, experience, and power positions that offer structural and sociological insights. Behaviour change scientists should recognise that their knowledge, including theories and experience, is not the only way of knowing and interpretations; participants from diverse backgrounds will bring about multiple epistemological ways in the co-designing process. Understanding this helps researchers stay humble, open, collaborative, and tolerant of differences brought up by different stakeholders. As such, to develop a behaviour change

intervention, it is crucial to view complex problems through both the lens of behaviour change theories and the lens of different co-designing participants, whose knowledge and experience of a specific area (e.g. local council's climate change planning) would be more extensive than that of the researchers.

In the Mission abovementioned, we utilised abductive reasoning to operate two important "lanes" when co-designing our interventions through a series of participatory workshops: the behaviour change lane and the knowledge co-production lane (see Figure 4). The behaviour change lane enabled us to approach the co-design procedure from a behavioural science perspective. We were able to conceptualise climate adaptation issues as behaviour challenges and employ theories of behaviour change, such as the Behaviour Change Wheel (Michie *et al.*, 2011, 2014) to guide our investigation of drivers and barriers for behaviour change. We utilised research tools such as evidence review to improve our grasp of shared concerns and applied measurement standards to complex issues of interest.

In knowledge co-production lane, we were open to any possibilities of knowledge emerging and contributing to our co-designing efforts, from mapping out climate adaptation challenges to behaviour identification and prioritisation, behaviour diagnosis and definition, charting out implementation context and adapting the intervention, to designing trial interventions and developing monitoring and evaluation procedures. We respected different or conflicting ideas raised in focus group discussions and used open-ended questions to explore different personal stories, experiences, and context-specific cases. We considered not only individuals' characteristics and positions in their communities and organisations but also their

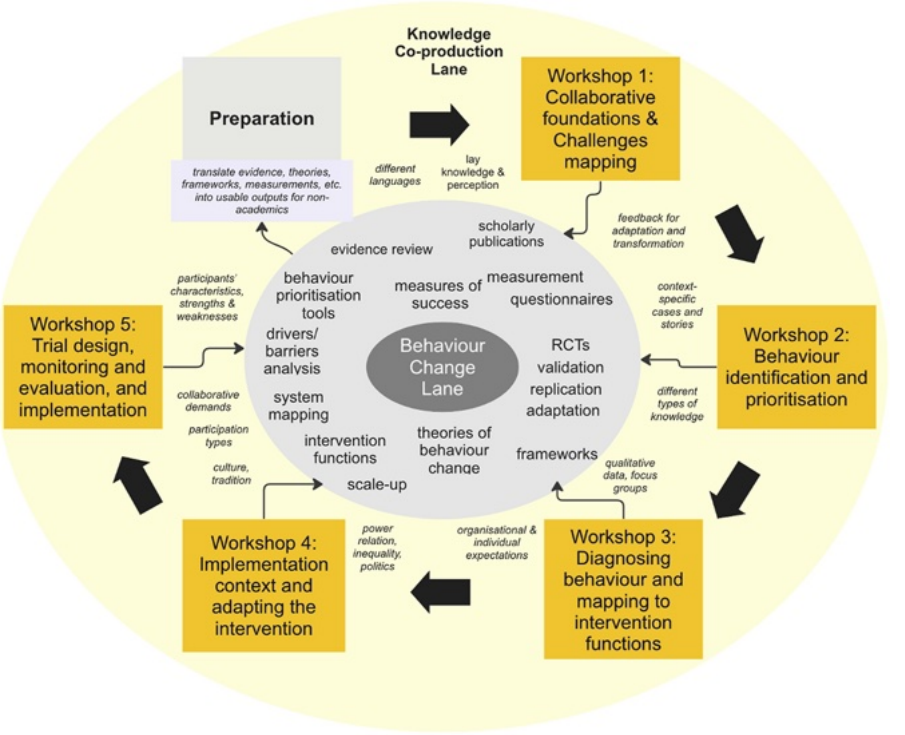


Figure 4.
Abductive integration of behavioural science and knowledge co-production

Source(s): Authors

community and organisational demands, different types of expectations, and participation capacity. We took into account the contextual differences in cultures, traditions, political powers, and power relations. With this combination, we were able to use systemic analyses, using theories like the socio-ecological model (Bronfenbrenner, 1977, 1979; Kerr *et al.*, 2014) or theory of practice (Shove, 2010) explain the influences of structural forces on behaviour change.

These two lanes were not separated but fluidly and dialectically blended. The behaviour change lens and its methods offered scientific evidence, theories, and frameworks that could illuminate the current problems of interest. To make them ready-to-use outputs, we needed to translate and transform them into accessible templates, questions, and evidence so that non-academic participants could understand. This translation task was important and taken into account in our preparation sessions before any workshop activities. For instance, in preparing for our co-design workshops, we used a systematic review to identify the climate adaptation challenges facing Australia in general and Victoria in particular. This systematic review provided evidence from both scholarly studies and grey literature about the current challenges. We identified over 750 challenges, grouped into 10 main challenges. However, in the co-design process, while informing the participants of these challenges, we were open to their judgement and choices about the climate challenges that made the most sense to them (Kaufman *et al.*, 2021). Through this process, participants were able to identify two key challenges for a desirable behaviour change intervention: one was the lack of farmers' engagement in natural capital markets, and another was that adaptation planning was not being led by Traditional Owners' knowledge.

In return, the co-design workshops offered rich insights into real-world problems, power relations, and cultures that helped (re)shape not only our design for these workshops but also our interpretation of the complex problems at hand.

Conclusion

In this article, we used collaborative and critical reflection (Fook and Gardner, 2007, 2012) and a pragmatic approach (Morgan, 2007) to document our lessons learnt in facilitating a two-year behaviour change project in Australia. The project was completed in June 2023 with much positive feedback from our partners, including government agencies, farmers' organisations, and Traditional Owners' associations (Saeri *et al.*, 2023). The success of this project can be attributed partly to the incorporation of collaborative reflection amongst the interdisciplinary team and partners. Our reflection was based on three stages of reflection, including *reflection in action* (reflexive and operational notes produced throughout the project), *reflection on knowledge co-production* (our purposeful reflection after the completion of two behaviour change interventions) and *critical reflection on the role of behavioural science in knowledge co-production* (our reflection sessions conducted before scaling-up workshops). These collaborative reflections involved multiple workshops and teamwork sessions in-person and online via Zoom and Miro Board to debunk our assumptions and routines in doing behavioural science projects and transform our approach to better support knowledge co-production in the context of climate change challenges.

Our reflection shows that to effectively support the process of knowledge co-production for climate adaptation challenges, the implementation of behaviour change projects should incorporate pragmatic principles that encourage inter- and transdisciplinary approaches, allow behavioural scientists to stay open, humble, collaborative, and give them space to exercise reflective practices throughout a project's cycle. To design a successful behaviour change project, researchers not only need individualistic lens from behavioural science, behavioural economic, psychology, which looks closely on individuals, their behaviours, capability, and motivation, but also need socio-cultural lens from sociology and anthropology and other paradigms (e.g. critical theory and political epistemologies), which encourages

holistic and structural analyses (Nguyen-Trung *et al.*, 2023). According to Morgan (2007, 2014), pragmatism provides an appropriate methodological foundation for engaging with various stakeholders in an open and collaborative manner while avoiding the constraints of (post)positivism and quantitative methods (Ewert, 2020; Kaufman *et al.*, 2021). Adopting a pragmatic approach is necessary for the behaviour change lens to increase its influence in policy-making, especially given the growing collaboration amongst different stakeholders and the need to decolonise studies by inviting more perspectives from local people from the Global South (Gaillard and Raju, 2022).

References

Adkins, L. (2003), "Reflexivity: freedom or habit of gender? Theory", *Culture and Society*, Vol. 20 No. 6, pp. 21-42, doi: 10.1177/0263276403206002.

Boulet, M., Grant, W., Hoek, A. and Raven, R. (2022), "Influencing across multiple levels: the positive effect of a school-based intervention on food waste and household behaviours", *Journal of Environmental Management*, Vol. 308, 114681, doi: 10.1016/j.jenvman.2022.114681.

Boulet, M., Hoek, A.C. and Raven, R. (2021), "Towards a multi-level framework of household food waste and consumer behaviour: untangling spaghetti soup", *Appetite*, Vol. 156, 104856, doi: 10.1016/j.appet.2020.104856.

Bourdieu, P. (1977), *Outline of a Theory of Practice*, Cambridge University Press, Cambridge.

Bourdieu, P. (1990), *The Logic of Practice*, Stanford University Press, Standford.

Braun, V. and Clarke, V. (2006), "Using thematic analysis in psychology", *Qualitative Research in Psychology*, Vol. 3 No. 2, pp. 77-101, doi: 10.1191/1478088706qp063oa.

Braun, V. and Clarke, V. (2019a), "Reflecting on reflexive thematic analysis", *Qualitative Research in Sport, Exercise and Health*, Vol. 11 No. 4, pp. 589-597, doi: 10.1080/2159676x.2019.1628806.

Braun, V. and Clarke, V. (2019b), "To saturate or not to saturate? Questioning data saturation as a useful concept for thematic analysis and sample-size rationales", *Qualitative Research in Sport, Exercise and Health*, Vol. 13 No. 2, pp. 1-16, doi: 10.1080/2159676x.2019.1704846.

Braun, V. and Clarke, V. (2021), "Can I use TA? Should I use TA? Should I not use TA? Comparing reflexive thematic analysis and other pattern-based qualitative analytic approaches", *Counselling and Psychotherapy Research*, Vol. 21 No. 1, pp. 37-47, doi: 10.1002/capr.12360.

Braun, V. and Clarke, V. (2022), "Conceptual and design thinking for thematic analysis", *Qualitative Psychology*, Vol. 9 No. 1, pp. 3-26, doi: 10.1037/qup0000196.

Brierley, J.A. (2017), "The role of a pragmatic paradigm when adopting mixed methods in behavioural accounting research", *International Journal of Behavioural Accounting and Finance*, Vol. 6 No. 2, p. 140, doi: 10.1504/ijbaf.2017.086432.

Bronfenbrenner, U. (1977), "Toward an experimental ecology of human development", *American Psychologist*, Vol. 32 No. 7, pp. 513-531, doi: 10.1037//0003-066x.32.7.513.

Bronfenbrenner, U. (1979), *The Ecology of Human Development: Experiments by Nature and Design*, Harvard University Press, New Delhi.

Bryan, C.J., Tipton, E. and Yeager, D.S. (2021), "Behavioural science is unlikely to change the world without a heterogeneity revolution", *Nature Human Behaviour*, Vol. 5 No. 8, pp. 1-10, doi: 10.1038/s41562-021-01143-3.

Cash, P.J., Hartlev, C.G. and Durazo, C.B. (2017), "Behavioural design: a process for integrating behaviour change and design", *Design Studies*, Vol. 48, pp. 96-128, doi: 10.1016/j.destud.2016.10.001.

Chater, N. and Loewenstein, G. (2022), "The i-frame and the s-frame: how focusing on individual-level solutions has led Behavioural science astray", *Behavioural and Brain Sciences*, pp. 1-60.

Ewert, B. (2020), "Moving beyond the obsession with nudging individual behaviour: towards a broader understanding of Behavioural science", *Public Policy and Administration*, Vol. 35 No. 3, pp. 337-360, doi: 10.1177/0952076719889090.

Fook, J. and Gardner, F. (2007), *Practising Critical Reflection: A Resource Handbook*, Open University Press, London.

Fook, J. and Gardner, F. (Eds). (2012), *Critical Reflection in Context: Applications in Health and Social Care*, Routledge, New York.

Gaillard, J.C. and Raju, E. (2022), "Editorial: on priorities, values and relationships in practice: a new road for disaster scholarly publishing", *Disaster Prevention and Management: An International Journal*, Vol. 31 No. 4, pp. 333-334, doi: 10.1108/dpm-08-2022-419.

Giddens, A. (1990), *The Consequences of Modernity*, Polity Press, Cambridge.

Grundy, E., Saeri, A., Kaufman, S., Klemm, C. (2021), Climate Adaptation Mission: Prioritisation Summit Briefing (report), Monash University, doi: 10.26180/23951049.v3.

Hallsworth, M. (2023), "A manifesto for applying behavioural science", *Nature Human Behaviour*, Vol. 7 No. 3, pp. 310-322, doi: 10.1038/s41562-023-01555-3.

Hansen, P.G. (2018), "What are we forgetting?", *Behavioural Science*, Vol. 2 No. 2, pp. 190-197, doi: 10.1017/bpp.2018.13.

John, P. and Stoker, G. (2019), "Rethinking the role of experts and expertise in behavioural science", *Policy and Politics*, Vol. 47 No. 2, pp. 209-225, doi: 10.1332/030557319X15526371698257.

Kaufman, S., Saeri, A., Raven, R., Malekpour, S. and Smith, L. (2021), "Behaviour in sustainability transitions: a mixed methods literature review", *Environmental Innovation and Societal Transitions*, Vol. 40, pp. 586-608, doi: 10.1016/j.eist.2021.10.010.

Kaufman, S., Slattery, P., Klemm, C. (2021), "Behaviour change in a changing climate: scoping paper in support of missions-based research and innovation for the BWA Consortium (report)", Monash University, doi: 10.26180/23961504.v1.

Kelly, L.M. and Cordeiro, M. (2020), "Three principles of pragmatism for research on organizational processes", *Methodological Innovations*, Vol. 13 No. 2, 2059799120937242, doi: 10.1177/2059799120937242.

Kerr, Z.Y., Register-Mihalik, J.K., Marshall, S.W., Evenson, K.R., Mihalik, J.P. and Guskiewicz, K.M. (2014), "Disclosure and non-disclosure of concussion and concussion symptoms in athletes: review and application of the socio-ecological framework", *Brain Injury*, Vol. 28 No. 8, pp. 1009-1021, doi: 10.3109/02699052.2014.904049.

Lamaison, P. and Bourdieu, P. (1986), "From rules to strategies: an interview with Pierre Bourdieu", *Cultural Anthropology*, Vol. 1 No. 1, pp. 110-120, doi: 10.1525/can.1986.1.1.02a00060.

Lash, S. (1994), "Expert-systems or situated interpretation? Culture and institutions in disorganized capitalism", in Beck, U., Giddens, A. and Lash, S. (Eds), *Reflexive Modernization: Politics, Tradition and Aesthetics in the Modern Social Order*, Polity Press, Cambridge, pp. 198-215.

Lash, S. (2003), "Reflexivity as non-linearity", *Theory, Culture and Society*, Vol. 20 No. 2, pp. 49-57, doi: 10.1177/0263276403020002003.

Lupton, D. and Watson, A.A. (2021), "Towards more-than-human digital data studies: developing research-creation methods", *Qualitative Research*, Vol. 21 No. 4, pp. 463-480, doi: 10.1177/1468794120939235.

Mazzucato, M. (2018), "Mission-oriented research and innovation in the European Union: a problem-solving approach to fuel innovation-led growth", *European Commission, Research and Innovation*, Vol. 36.

Michie, S., van Stralen, M.M. and West, R. (2011), "The behaviour change Wheel: a new method for characterising and designing behaviour change interventions", *Implementation Science*, Vol. 6 No. 1, p. 42, doi: 10.1186/1748-5908-6-42.

Michie, S., Atkins, L. and West, R. (2014), *The Behaviour Change Wheel: A Guide to Designing Interventions*, Silverback, Thousand Oaks.

Miro. (2024), About Miro, available at: https://miro.com/about/

Morgan, D.L. (2007), "Paradigms lost and pragmatism regained: methodological implications of combining qualitative and quantitative methods", *Journal of Mixed Methods Research*, Vol. 1 No. 1, pp. 48-76, doi: 10.1177/2345678906292462.

Morgan, D.L. (2014), "Pragmatism as a paradigm for social research", *Qualitative Inquiry*, Vol. 20 No. 8, pp. 1045-1053, doi: 10.1177/1077800413513733.

Morgan, D. (2020), "Pragmatism as a basis for grounded theory", *The Qualitative Report*, Vol. 25 No. 1, p. 64, doi: 10.46743/2160-3715/2020.3993.

Nguyen-Trung, K. (2022), "Risk epistemologies and aesthetic reflexivity of a disaster-affected community: findings from Vietnam", *Sociological Research Online*, Vol. 27 No. 4, pp. 932-946, doi: 10.1177/13607804221133120.

Nguyen-Trung, K. (2023), "Reflexivity, habitus and vulnerability: vietnamese farmers' attribution of responsibility in a post-disaster context", *Disaster Prevention and Management: An International Journal*, Vol. 32 No. 2, pp. 298-310, doi: 10.1108/dpm-05-2022-0118.

Nguyen-Trung, K., Saeri, A.K., Zhao, K., Boulet, M. and Kaufman, S. (2023), "A brief introduction to a socio-ecological COM-B (SeCOM-B): a behaviour change framework response to wicked problems", available at: https://org/10.31219/osf.io/4x6wa

Norström, A.V., Cvitanovic, C., Löf, M.F., West, S., Wyborn, C., Balvanera, P., Bednarek, A.T., Bennett, E.M., Biggs, R., de Bremond, A., Campbell, B.M., Canadell, J.G., Carpenter, S.R., Folke, C., Fulton, E.A., Gaffney, O., Gelcich, S., Jouffray, J.-B., Leach, M., Le Tissier, M., Martín-López, B., Louder, E., Loutre, M.F., Meadow, A.M., Nagendra, H., Payne, D., Peterson, G.D., Reyers, B., Scholes, R., Speranza, C.I., Spierenburg, M., Stafford-Smith, M., Tengö, M., van der Hel, S., van Putten, I., Österblom, H. and Österblom, H. (2020), "Lessons learnt for knowledge co-production in sustainability research", *Nature Sustainability*, Vol. 3 No. 3, pp. 182-190, doi: 10.1038/s41893-019-0448-2.

Reynolds, M. (1998), "Reflection and critical reflection in management learning", *Management Learning*, Vol. 29 No. 2, pp. 183-200.

Rittel, H.W. and Webber, M.M. (1973), "Dilemmas in a general theory of planning", *Policy Sciences*, Vol. 4 No. 2, pp. 155-169.

Saeri, A., Nguyen Trung, K. and Kaufman, S. (2023), *Trial Report: DJAARA-Led Adaptation Forum for Local Governments*, Monash Sustainable Development Institute, Monash University, BehaviourWorks Australia.

Schimmelpfennig, R. and Muthukrishna, M. (2023), "Cultural evolutionary behavioural science in public policy", *Behavioural Science*, pp. 1-31, doi: 10.1017/bpp.2022.40.

Shove, E. (2010), "Beyond the ABC: climate change policy and theories of social change", *Environment and Planning A: Economy and Space*, Vol. 42 No. 6, pp. 1273-1285, doi: 10.1068/a42282.

Straßheim, H. (2020), "The rise and spread of behavioural science: an opportunity for critical research and self-reflection", *International Review of Public Policy*, Vol. 2 No. 1, pp. 115-212.

Thaler, R.H. and Sunstein, C.R. (2008), *Nudge: Improving Decisions about Health, Wealth, and Happiness*, Yale University Press, New Heavan.

Zoom Video Communications (2024), About Zoom, available at: https://www.zoom.com/en/about/

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